

Archived results (not part of the peer-reviewed paper)

Yi Wang^{1,*} and Yongqing Tang²

¹ College of Engineering, Lishui University, Lishui 323000, Zhejiang, China

² School of Physics, Mechanics & Electronics, Longyan University, Longyan, Fujian, China

E-mail: jefflsxy@lsu.edu.cn

This document archives results produced during the study “When does a second vibration sensor help? A controlled case study of sensor configuration for deep-learning bearing fault diagnosis” that are *not* part of the peer-reviewed paper or of its supplementary material, because they do not bear on the sensor-configuration question: comparisons of the three learners with one another, with further baselines and with normalisation variants (section A2); a forward-pass latency benchmark (section A3); and an exploratory kinematic frequency-alignment regulariser for the Mamba-3 model, which is not a contribution of the study (sections A4 and A5). No claim of the paper rests on this document. Sections, tables, figures and equations carry the prefix A; “supplementary section/table S_n ” refers to the peer-reviewed supplementary material, and “section 3.x/4.x/5” to the main text, whose numbers and wording take precedence. All values were produced under the leakage-free protocol (supplementary section S4) unless stated otherwise.

Naming. The three learners are the Mamba-3 model, the Mamba-2 model and the 1D-CNN; DE, FE and DE+FE denote CWRU drive-end, fan-end and dual input; “CE only” denotes training with the cross-entropy loss alone and “ $+\mathcal{L}_{\text{kin}}$ ” training with the regulariser of section A4. Several figures were re-rendered from their original plotting scripts and result files with revised titles and legends only. Four t-SNE figures (figures A8, A9, A10 and A11) could not be re-rendered without changing the plotted coordinates, so their titles still use the earlier name “BearMamba-3” for the Mamba-3 model.

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A1. Results obtained with the superseded noise implementation

Superseded noise protocol; not used to support the current manuscript’s conclusions. The CWRU and Paderborn loaders used for every noise-injected result in this archive seeded the injected noise by a window’s index within its split, so training, validation and test windows with the same index received the same standard-normal noise waveform, and the training seed re-paired test windows with noise. The raw signals of the splits were disjoint. The current manuscript uses only Mamba-3 CWRU results obtained with a corrected noise implementation (manuscript section 3.5, item 9). All noise-injected CWRU and Paderborn results in this archive, including the learner comparisons and the kinematic-regulariser results, belong to the superseded protocol; noise-free results are unaffected. The blocks below were moved here from the peer-reviewed supplement unchanged apart from heading levels; favourable and unfavourable results are kept alike.

A1.1. Channel coherence: CWRU noise sweep and gain-coherence correlation (former supplementary section S1.1)

This section supports the paragraph “Channel coherence” of section 4.4. The estimator and data handling are described in supplementary section S1.2. The analysis is read-only: it uses the raw recordings and trains no model.

A1.1.1. Coherence and gain

Baseline coherence. Without injected noise, the magnitude-squared coherence between the two channels in the analytically derived fault-frequency band is moderate on both datasets (CWRU, DE and FE: mean 0.412, range 0.307–0.492; XJTU-SY, H and V: mean 0.417, range 0.337–0.554). The two channels therefore share part, but not all, of their fault-band content. As a sanity check, the per-channel fault-band SNR is higher in faulted than in fault-free CWRU cells (2.65 against 0.62 dB).

CWRU noise sweep. Injecting independent AWGN into each channel (section 3.3) lowers the fault-band coherence by construction, because the independent noise increasingly dominates the shared fault signal. Measured coherence fell monotonically as SNR decreased (0 dB: 0.013; −2 dB: 0.008; −4 dB: 0.004; −6 dB: 0.001; −8 dB: 0.001), while the Mamba-3 DE+FE minus DE gain (CE only; table A4) rose from +0.03 through +0.42, +0.59 and +0.62 to +2.71 pp. The Spearman correlation over these five points is $r = -1.00$ (exact permutation $p = 1/60$). Both quantities are driven by the same injected noise, so this co-variation is consistent with a spatial-averaging account but is not independent evidence for it. With $+\mathcal{L}_{\text{kin}}$ (archive, section A3) the within-CWRU relation was not significant ($r = -0.67$, $p = 0.27$).

XJTU-SY. The XJTU-SY splits involve no injected noise. The LOBO and cross-condition settings have nearly the same coherence (0.412 and 0.429) but gains of opposite sign (-18.5 pp LOBO, $+26.6$ pp cross-condition; both first software environment, CE only, $n = 8$). Pooling all seven points (five CWRU, two XJTU-SY) gives $r = -0.25$ (exact permutation $p = 0.59$). In these data, coherence did not predict the sign or the magnitude of the gain, and the LOBO/cross-condition sign difference is not explained by it. All seven points are listed in table A1 and plotted in figure A1, so that the unfavourable points are reported alongside the CWRU sweep.

Table A1. Fault-band channel coherence and second-sensor gain (dual minus reference single sensor, pp) for all seven points of the coherence analysis, leakage-free protocol, CE only. CWRU: DE+FE minus DE, Mamba-3 model, $n = 8$ seeds at -4 , -6 and -8 dB and $n = 5$ at 0 and -2 dB (table A4). XJTU-SY: H+V minus H, Mamba-3 model, $n = 8$, first software environment (supplementary table S15).

Dataset	Condition	Coherence	Gain (pp)
CWRU	SNR = 0 dB	0.0133	+0.03
CWRU	SNR = -2 dB	0.0082	+0.42
CWRU	SNR = -4 dB	0.0039	+0.59
CWRU	SNR = -6 dB	0.0015	+0.62
CWRU	SNR = -8 dB	0.0009	+2.71
XJTU-SY	LOBO, leakage-free	0.4117	-18.5
XJTU-SY	Cross-condition (condition 2 $\rightarrow$ 3)	0.4285	$+26.6$

A1.2. Pink-noise condition (former supplementary section S3)

A1.2.1. Pink-noise condition: generator spectrum, in-band SNR and difficulty This section supports the paragraph “Noise colour” of section 4.4. The CWRU DE and DE+FE Mamba-3 cells at -4 , -6 and -8 dB (CE only) were repeated with pink ($1/f$) noise in place of AWGN at the same nominal broadband SNR. Each channel’s noise realisation is drawn independently.

Generator check. Figure A2 verifies that the released pink-noise generator produces $1/f$ -shaped noise: the fitted log–log slope of the power spectral density is -0.998 for a pink realisation and $+0.000$ for a white-noise control of the same length.

Results. Under pink noise DE+FE minus DE was -0.21 , -0.45 and -0.39 pp (table A2; $n = 5$, two-sided Wilcoxon, $p \geq 0.19$). A positive gain was not detected, and a loss was not shown.

Difficulty and in-band SNR. The pink and white cells are not matched in difficulty. At the same nominal broadband SNR, DE accuracy is 97.58% under pink noise against 85.76% under AWGN at -8 dB (and 99.73% against 97.12% at -4 dB), so the pink cells

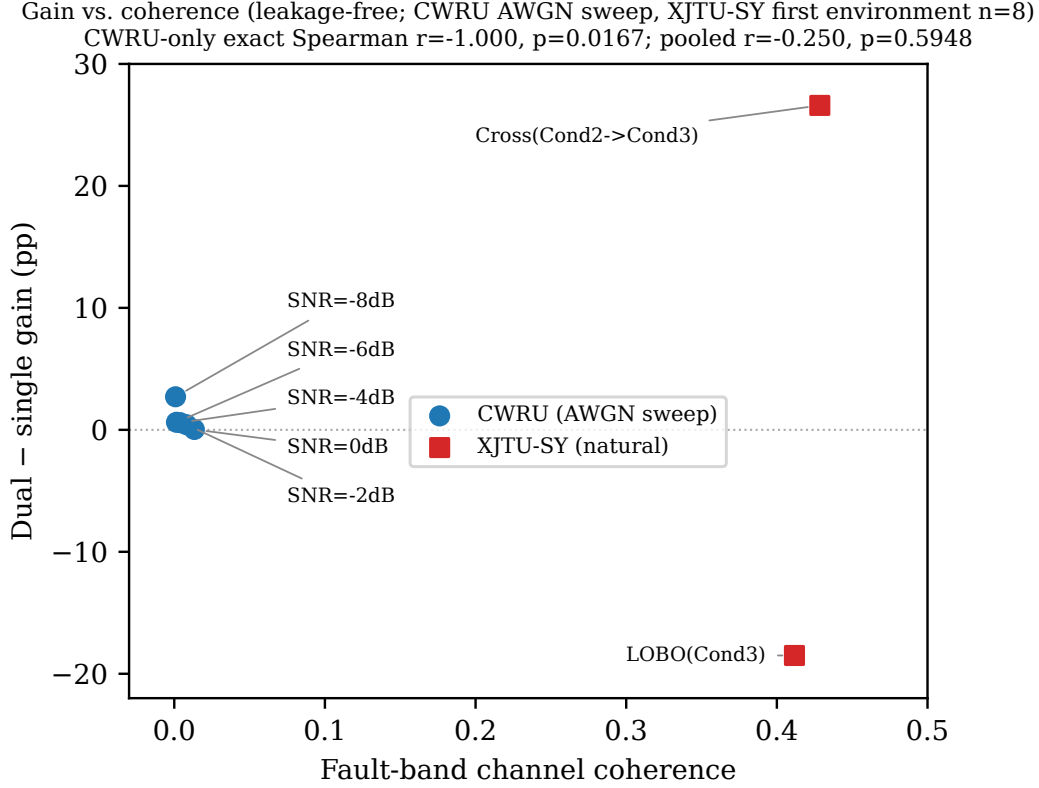


Figure A1. Second-sensor gain against fault-band coherence, all seven points of table A1, leakage-free protocol. CWRU only: exact Spearman $r = -1.00$, permutation $p = 1/60$; pooled (7 points): $r = -0.25$, $p = 0.59$. Within one dataset both quantities are monotone in SNR, so the CWRU-only correlation follows from the shared driver (the injected noise both lowers coherence and raises the gain) and is not independent evidence for a mechanism. The XJTU-SY points are the $n = 8$ first-environment values of supplementary table S15; “dual – single” on the vertical axis is DE+FE – DE (CWRU) or H+V – H (XJTU-SY).

Table A2. CWRU DE and DE+FE accuracy (%) under pink ($1/f$) noise, Mamba-3 model, CE only, leakage-free protocol, mean $\pm$ SD over 5 seeds; two-sided exact Wilcoxon p for DE+FE minus DE. No cell is significant.

SNR	DE	DE+FE	Δ (DE+FE – DE)	p
–4 dB	99.73 $\pm$ 0.29	99.52 $\pm$ 0.22	–0.21	0.2500
–6 dB	99.21 $\pm$ 0.35	98.76 $\pm$ 0.46	–0.45	0.1875
–8 dB	97.58 $\pm$ 1.55	97.18 $\pm$ 1.00	–0.39	0.6250

Pink vs white noise PSD — spectral verification of the pink-noise generator

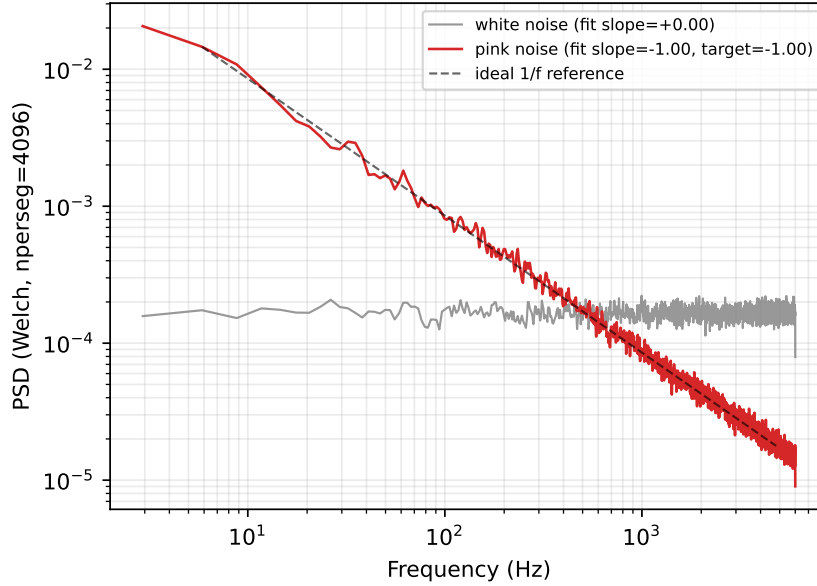


Figure A2. Log-log power spectral density of a pink-noise realisation (fitted slope -0.998 ; ideal $1/f$ noise: -1.0) and of a white-noise control of the same length (fitted slope $+0.000$; expected 0.0). Both fits pass the in-script assertion tolerance. Generated by `experiments/exp_e5_pink_noise/verify_pink_spectrum.py` (repository path; re-rendered for this version with a revised title only, same seed and fit).

lie near the ceiling, where the largest attainable gain is $100 - \text{DE} \approx 2.4$ pp at -8 dB and 0.3 pp at -4 dB. Table A3 shows the spectral reason: with power $\propto 1/f$, pink noise at equal broadband SNR places about 22% of its power in the 30–170 Hz kinematic band (AWGN: 2%) but leaves the 2–5 kHz band, where bearing resonances and impact transients appear, at a higher SNR ($+0.1$ dB against -5.8 dB at nominal -8 dB). The classifier scores higher under pink noise despite the lower SNR in the kinematic band, so nominal broadband SNR is not a difficulty-matched axis. The account that pink noise overlaps the fault band and so cancels the benefit of averaging predicts the opposite accuracy ordering from the one observed and is not supported.

Table A3. Median per-window SNR in two bands and share of noise power in the 30–170 Hz band, for AWGN and pink noise injected at the same nominal broadband SNR (CWRU DE channel, 12 kHz, 2048-sample windows after mean removal; 40 random windows from each of four recordings, normal, IR, ball and OR at 0.007 inch; 160 windows per row).

Noise	Nominal SNR	SNR 30–170 Hz (dB)	SNR 2–5 kHz (dB)	Noise share 30–170 Hz
AWGN	-4 dB	-13.7	-1.7	0.023
Pink	-4 dB	-23.3	$+4.1$	0.225
AWGN	-8 dB	-17.9	-5.8	0.023
Pink	-8 dB	-27.2	$+0.1$	0.223

Comparison at matched DE accuracy. AWGN at -4 dB (DE 97.37%, gain +0.59 pp, $n = 8$) and pink noise at -8 dB (DE 97.58%, gain -0.39 pp, $n = 5$) have similar single-sensor accuracy. The difference in gain is 0.98 pp with a bootstrap 95% interval of -0.28 to $+2.33$ pp (Mann–Whitney $p = 0.35$), which does not separate the two noise types. Averaged over the three SNRs on the shared seeds 0–4, the AWGN gain exceeds the pink gain by 2.17 pp (interval 0.67 to 3.92, Mann–Whitney $p = 0.032$), but that comparison is confounded with difficulty. A positive gain was therefore not detected under pink noise, and whether the gain depends on noise colour is unresolved. A decisive test would inject pink and white noise at matched single-sensor accuracy or matched in-band SNR; it was not run (section 5, “Missing controls”), and a dependence on noise colour remains a hypothesis.

A1.3. CWRU per-SNR results (former supplementary section S6.7)

A1.3.1. CWRU per-SNR results This subsection supports section 4.4 and table 5 of the main text. The noise-colour and coherence analyses are in the preceding subsections of this section.

DE against DE+FE, Mamba-3 learner. Figure A3 and table A4 compare DE and DE+FE across the SNR grid (first software environment). The -4 , -6 and -8 dB cells use 8 seeds: the original five plus three added after a re-check found that the originally reported -4 dB result ($p = 0.031$, $n = 5$) did not hold with additional seeds (section 3.5, item “Seed extensions”); the 0 and -2 dB cells remain at $n = 5$. The $+10$ dB point is excluded because DE accuracy is already 100.00%. The DE+FE minus DE difference (CE only) is positive at every SNR and largest at -8 dB (+0.59, +0.62 and +2.71 pp at -4 , -6 and -8 dB). This direction is consistent with a spatial-averaging account (supplementary section S5.1). At the one-sided $\alpha = 0.05$ of the declared directional hypothesis, two CE-only cells of table A4 have $p < 0.05$: -8 dB ($p = 0.0039$, $n = 8$, the $n = 8$ floor) and -2 dB ($p = 0.0312$, $n = 5$, the $n = 5$ floor); with $+\mathcal{L}_{\text{kin}}$ the -8 dB cell also has $p < 0.05$ ($p = 0.0391$, $n = 8$). The -4 , -6 and -8 dB cells have $n = 8$ because they were extended post hoc from five seeds after the originally reported -4 dB result did not hold; the 0 and -2 dB cells have $n = 5$. No correction for multiple comparisons was applied across the five SNRs or the two training arms, so all of these are nominal, unadjusted p values, and the -8 dB cell, analysed after both the seed extension and the protocol correction, is exploratory. The CWRU AWGN result is therefore descriptive.

Fan-end-only control. FE-only models were trained at -4 , -6 and -8 dB with the same configuration (5 seeds, leakage-free; table A5). FE alone was weaker than DE at every SNR and for both learners (3.4–10.3 pp, all 30 seed-paired comparisons in the same direction), so the DE+FE gain is not accounted for by the added channel being the better one. For the Mamba-3 learner, adding FE to DE raised accuracy by +1.2, +1.5 and +2.8 pp (bootstrap intervals $[0.5, 2.2]$, $[-0.3, 3.2]$ and $[0.6, 5.5]$; the exact two-sided

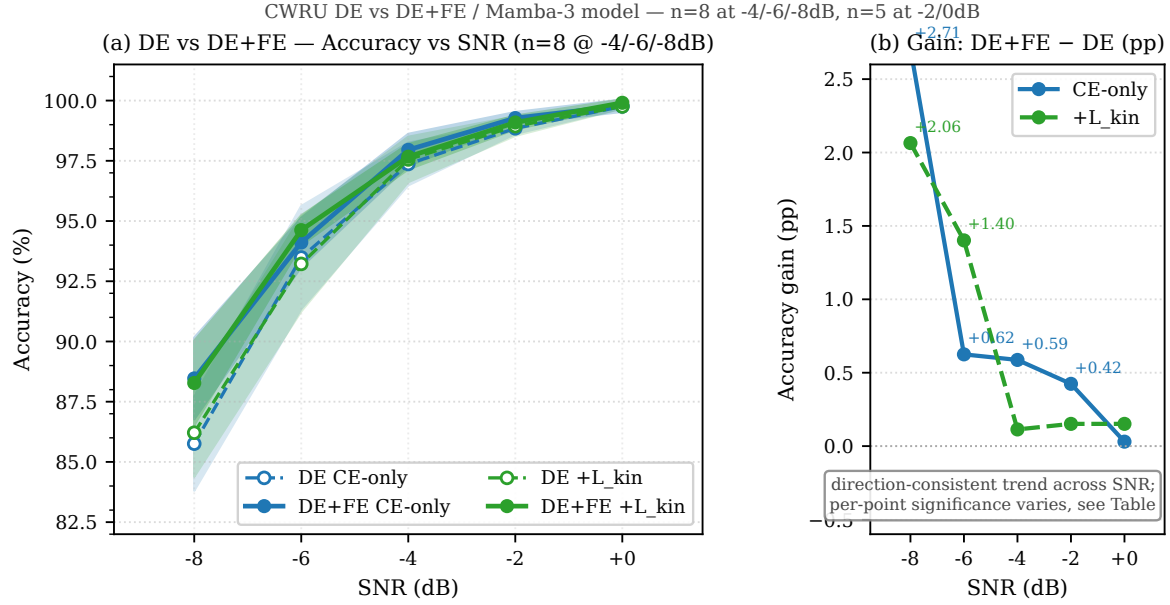


Figure A3. CWRU DE and DE+FE accuracy of the Mamba-3 model against SNR (± 1 SD shaded; $n = 8$ seeds at -4 , -6 and -8 dB, $n = 5$ at -2 and 0 dB; first software environment). (a) Accuracy: solid, DE+FE; dashed, DE; blue, CE only; green, $+\mathcal{L}_{\text{kin}}$. (b) DE+FE minus DE (pp); per-point p values (nominal, one-sided, not adjusted for multiplicity) are in table A4.

Table A4. CWRU DE and DE+FE test-segment accuracy (%), Mamba-3 model, first software environment, leakage-free protocol, mean $\pm$ SD. The -4 , -6 and -8 dB cells use the merged 8-seed pool (original 5 seeds plus 3 added seeds, both training arms); -2 and 0 dB remain $n = 5$. p : one-sided exact Wilcoxon for DE+FE $>$ DE, the directional hypothesis declared before analysis (section 3.4); nominal values, no multiplicity correction.

SNR	n	DE, CE only	DE, $+\mathcal{L}_{\text{kin}}$	DE+FE, CE only	Δ , CE only	p , CE only	DE+FE, $+\mathcal{L}_{\text{kin}}$	p , $+\mathcal{L}_{\text{kin}}$
-8 dB	8	85.76 ± 1.95	86.21 ± 1.83	88.47 ± 1.69	$+2.71$	0.0039	88.28 ± 1.75	0.0391
-6 dB	8	93.48 ± 2.13	93.22 ± 1.97	94.11 ± 1.02	$+0.62$	0.3203	94.62 ± 0.62	0.0547
-4 dB	8	97.37 ± 0.88	97.56 ± 0.94	97.95 ± 0.66	$+0.59$	0.1328	97.67 ± 0.52	0.5234
-2 dB	5	98.85 ± 0.23	98.94 ± 0.40	99.27 ± 0.25	$+0.42$	0.0312	99.09 ± 0.28	0.4375
0 dB	5	99.76 ± 0.08	99.76 ± 0.08	99.79 ± 0.25	$+0.03$	0.4375	99.91 ± 0.14	0.1250

The -4 dB result originally reported as significant ($+0.70$ pp, $p = 0.031$, $n = 5$, earlier protocol) does not hold ($+0.59$ pp, $p = 0.13$, $n = 8$). At -8 dB all eight CE-only seeds are concordant ($p = 0.0039$, the $n = 8$ floor); because this cell was analysed after both the seed extension and the protocol correction, it is reported as exploratory, not confirmatory. With $+\mathcal{L}_{\text{kin}}$ the one-sided p values are ≥ 0.039 at every SNR; this is reported without interpretation.

Wilcoxon floor at $n = 5$ is 0.0625, so none is significant at 0.05). For the 1D-CNN, which is already above 99% with DE at -4 and -6 dB, the gains were $+0.1$, $+0.7$ and $+4.3$ pp. The better single sensor was DE in every case, so DE+FE minus the better single sensor equals DE+FE minus DE. The Mamba-3 DE and DE+FE values in this paragraph and in table A5 come from the first-environment result files; the main text (table 5) now reports Mamba-3 DE and DE+FE arms retrained in the second environment ($+0.5$, $+1.9$ and $+3.3$ pp; per seed in table A6). The 1D-CNN arms were all trained in the second environment (section 3.5). The Mamba-3 values here ($n = 5$, seeds 0–4) differ from the 8-seed pool of table A4 at -4 , -6 and -8 dB because the seed sets differ.

Table A5. CWRU accuracy (%) with FE, DE and DE+FE input under AWGN, mean $\pm$ SD over 5 seeds (0–4). Mamba-3 DE and DE+FE: original leakage-free result files (first software environment); FE and all 1D-CNN rows: second software environment.

Learner	SNR	FE	DE	DE+FE
Mamba-3	-4 dB	92.55 ± 0.91	97.12 ± 0.99	98.33 ± 0.47
	-6 dB	84.42 ± 0.96	92.67 ± 1.79	94.12 ± 1.13
	-8 dB	76.09 ± 2.20	85.09 ± 1.76	87.88 ± 1.80
1D-CNN	-4 dB	96.48 ± 0.91	99.85 ± 0.11	99.97 ± 0.07
	-6 dB	91.76 ± 0.93	99.18 ± 0.14	99.91 ± 0.08
	-8 dB	84.18 ± 1.52	94.45 ± 0.79	98.73 ± 0.35

A1.4. Round-3 CWRU per-seed table (former table S36)

Table A6. CWRU, Mamba-3, environment 2: test accuracy (%) at the validation-selected epoch per seed. FE-only: arms of supplementary section S6; all other arms trained for this revision. ρ : correlation of the injected noise between the two channels (no ρ : independent noise).

Arm	seed 0	seed 1	seed 2	seed 3	seed 4
-4 dB, FE	91.67	93.18	93.79	92.27	91.82
-4 dB, DE	96.21	98.03	98.48	97.27	95.76
-4 dB, DE+FE	97.88	97.42	98.33	97.58	97.12
-6 dB, FE	85.15	84.09	85.00	82.88	85.00
-6 dB, DE	90.00	91.82	93.48	93.03	93.79
-6 dB, DE+FE	94.85	95.61	94.24	93.33	93.48
-6 dB, DE+FE, $\rho = 0.5$	96.67	96.82	96.21	93.94	95.91
-6 dB, DE+FE, $\rho = 0.9$	99.24	99.70	99.24	97.88	99.09
-8 dB, FE	77.42	73.64	73.94	76.82	78.64
-8 dB, DE	84.24	84.70	83.94	85.15	85.76
-8 dB, DE+FE	87.42	88.64	88.79	86.97	88.33
-8 dB, DE+FE, $\rho = 0.5$	90.30	90.45	90.45	90.91	90.76
-8 dB, DE+FE, $\rho = 0.9$	99.09	97.73	97.58	98.18	96.67

A2. Comparisons of the learners with one another

This section contains the learner-comparison parts of the former supplementary section on the learners. The learners are tools for the sensor-configuration question of the paper; the comparisons reported here are descriptive, no difference between learners is attributed to a mechanism, and no learner is claimed to be generally more accurate. The architecture, classification head, embedding and hyperparameters of the three learners are in supplementary section S7.

A2.1. Paderborn dataset

Accelerometer measurements [1] from an SKF 6203 bearing ($n_b = 8$, $D = 29.05$ mm, $f_s = 64$ kHz) under several speed-load conditions. Three classes: normal (K001, K002), OR (KA04, KA15) and IR (KI01, KI03, KI05), excluding run-to-failure bearings (KB23, KB24, KB27) and ball bearings. The cross-condition protocol trains on N09_M07_F10 (900 rpm) and N15_M01_F10 (1500 rpm) and tests on N15_M07_F10 (1500 rpm, higher load), with a file-level split. Fault frequencies at 1500 rpm: $f_{\text{BPFO}} = 76.77$ Hz, $f_{\text{BPFI}} = 123.23$ Hz. The Paderborn dataset is used only in this document.

A2.2. Further CWRU baselines

Three methods were run on the CWRU four-class DE task with the same data pipeline, leakage-free splits and seed count (5 seeds; SK-SVM and 1D-CNN 50 epochs, Transformer-1D 20 epochs), with the AdamW learning rate of the Mamba-3 model (3×10^{-4} , supplementary table S27) and the protocol of the BatchNorm ablation (table A10). No per-baseline hyperparameter search was run, so the baselines are not tuned independently.

- *SK-SVM*: spectral-kurtosis feature vector (RMS, kurtosis, skewness, crest factor, envelope kurtosis and five spectral moments) and an RBF-kernel SVM ($C = 10$, $\gamma = \text{scale}$) [2]; scikit-learn pipeline, no GPU. Its hand-designed frequency features would have to be recomputed per sample under variable shaft speed.
- *1D-CNN*: the 1D-CNN learner of the main text (architecture: supplementary section S7.2); 100,676 parameters (four-class).
- *Transformer-1D*: four pre-norm Transformer encoder layers ($d_{\text{model}} = 64$, 4 heads, $d_{\text{ff}} = 256$, dropout 0.1), the same convolutional embedding, global average pooling; 200,900 parameters; no patch tokenisation.

CWRU is close to ceiling: the deep models differ little without noise and at high SNR, and the differences grow at low SNR (table A7); the cause of these differences was not identified. The Mamba-3 model is not claimed to be more accurate than the baselines on CWRU. Section A2.4 reports that the 1D-CNN’s within-condition advantage disappears when BatchNorm is removed and returns with `GroupNorm(1,`

Table A7. CWRU four-class DE accuracy (%) at representative noise levels, leakage-free protocol, mean $\pm$ SD over 5 seeds (0–4; first software environment). Clean: no injected noise. +10 dB: the +10 dB AWGN grid point. Parameters (four-class): SK-SVM, no GPU; 1D-CNN 100,676; Transformer-1D 200,900; Mamba-2 model 173,980; Mamba-3 model 178,068.

Method	Clean	+10 dB	−4 dB	−8 dB
SK-SVM	99.39 $\pm$ 0.00	—	86.12 $\pm$ 0.74	68.45 $\pm$ 1.54
1D-CNN	100.00 $\pm$ 0.00	—	99.88 $\pm$ 0.13	94.64 $\pm$ 1.31
Transformer-1D	99.88 $\pm$ 0.07	—	95.82 $\pm$ 0.79	82.48 $\pm$ 0.71
Mamba-2 model	n/a ^a	99.97 $\pm$ 0.07	98.21 $\pm$ 1.13	92.03 $\pm$ 1.10
Mamba-3 model, CE only	99.88 $\pm$ 0.13	99.94 $\pm$ 0.14	97.12 $\pm$ 0.99	85.09 $\pm$ 1.76
Mamba-3 model, + $\mathcal{L}_{\text{kin}}$	99.91 $\pm$ 0.08	99.91 $\pm$ 0.08	97.24 $\pm$ 1.07	85.94 $\pm$ 1.54

^a No no-noise Mamba-2 run exists under the leakage-free protocol; the Mamba-2 value previously listed in a “Clean” column is the +10 dB grid point (`noise_snr_db`= 10) and is now in the +10 dB column. —: not part of this table. The +10 dB values of the Mamba-3 model are those of table A16.

The −4 dB Mamba-3 CE-only value (97.12) is the $n = 5$ value also used in tables A16 and A10; the $n = 8$ DE value of supplementary table S23 (97.37 $\pm$ 0.88) is a different seed pool, not a different result.

d), and that the Mamba-3 model scored higher than the 1D-CNN on one single-sensor cross-condition split (+22.9 pp, $n = 8$).

A2.3. Mamba-3 and Mamba-2 models

Paderborn cross-condition. Table A8 compares the Mamba-3 model (SISO, CE only and + $\mathcal{L}_{\text{kin}}$) with the Mamba-2 model (CE only). Neither paired difference excludes zero (95% CI; Wilcoxon $p = 0.8125$ CE only, $p = 0.3125$ + $\mathcal{L}_{\text{kin}}$). With $n = 5$ seeds and no pre-specified equivalence margin or two one-sided tests (TOST), this is reported as no detected difference, not as equivalence or non-inferiority in the formal sense. The inter-seed SD of the Mamba-3 CE-only arm is about 7 $\times$ that of the Mamba-2 model (0.83 against 0.12); the reason was not investigated.

Table A8. Paderborn cross-condition accuracy (%), leakage-free protocol (source-domain validation split; test condition evaluated once), mean $\pm$ SD over 5 seeds, with the paired mean difference to the Mamba-2 model and its 95% CI (paired t interval, $n = 5$). A Mamba-2 + $\mathcal{L}_{\text{kin}}$ arm is architecturally undefined.

Learner	Accuracy (%)	Δ vs Mamba-2 (pp)	95% CI (pp)
Mamba-3 model, CE only	95.19 $\pm$ 0.83	+0.22	[−0.75, +1.19]
Mamba-3 model, + $\mathcal{L}_{\text{kin}}$ ($\lambda = 0.01$)	95.49 $\pm$ 0.76	+0.51	[−0.36, +1.38]
Mamba-2 model, CE only	94.97 $\pm$ 0.12	—	—

CWRU AWGN, DE. Under CWRU AWGN (table A16) the Mamba-2 model scored higher than the Mamba-3 model at -6 and -8 dB (two-sided $p = 0.063$ each, 5/5 seeds); at -4 dB the difference has the same sign but $p = 0.19$, so no statement is extended to that point. The reason for the difference was not investigated. Among the learners used here only the Mamba-3 model has complex oscillatory states, so only it can carry the frequency-alignment regulariser of section A4; this is a structural property, not an accuracy result.

Dual-sensor comparison by regime. Before these runs, the experiment log fixed how the comparison of the dual-sensor Mamba-2 and Mamba-3 models would be reported: if the Mamba-2 model matched or exceeded the Mamba-3 model in both the CWRU low-SNR and the XJTU-SY cross-condition regime, the learner comparison would be reported as regime-independent; otherwise it would be reported per regime. The first condition was not met (table A9), so the results are reported per regime. Under heavy CWRU noise the DE+FE Mamba-2 model was more accurate than the DE+FE Mamba-3 model by $+0.3$ to $+8.8$ pp (5/5 seeds at each of -4 , -6 and -8 dB); on XJTU-SY condition 2 $\rightarrow$ 3 the H+V Mamba-3 model was more accurate than the H+V Mamba-2 model by $+4.5$ to $+40.1$ pp (8/8 seeds, exact two-sided $p = 0.0078$; mean difference $+19.1$ pp, 89.72% against 70.63%). On the same split the H+V Mamba-3 macro-F1 was 44.2 pp above that of the H+V 1D-CNN (89.72% against 45.50%). Each regime is one dataset and, for XJTU-SY, one condition pair; neither result is extended to a deployment rule, and neither model is claimed to be generally more accurate.

Table A9. Dual-sensor Mamba-2 and Mamba-3 models by regime, leakage-free protocol, CE only, mean $\pm$ SD (CWRU rows: DE+FE, $n = 5$, exact Wilcoxon $p = 0.063$, 5/5 concordant; XJTU-SY row: H+V, $n = 8$, exact Wilcoxon $p = 0.0078$, 8/8 concordant; first software environment; descriptive). CWRU rows are within-condition comparisons.

Regime	Metric	Mamba-2	Mamba-3	Higher (range, pp) ^a
CWRU, AWGN -4 dB	acc. (%)	99.58 ± 0.35	98.33 ± 0.47	Mamba-2 ($+0.3$ to $+1.7$)
CWRU, AWGN -6 dB	acc. (%)	98.61 ± 0.46	94.12 ± 1.13	Mamba-2 ($+3.5$ to $+6.4$)
CWRU, AWGN -8 dB	acc. (%)	94.82 ± 1.01	87.88 ± 1.80	Mamba-2 ($+5.3$ to $+8.8$)
XJTU-SY cond. 2 $\rightarrow$ 3	macro-F1 (%)	70.63 ± 9.68	89.72 ± 9.98	Mamba-3 ($+4.5$ to $+40.1$)

^a Range of the per-seed paired differences (not the difference of means); concordance is 5/5 (CWRU) or 8/8 (XJTU-SY).

A2.4. Normalisation ablation and cross-condition comparison with the 1D-CNN

This subsection compares the Mamba-3 model with the 1D-CNN within a condition (CWRU) and across conditions (XJTU-SY), and reports the effect of changing the 1D-CNN’s normalisation. The comparison is observational: the models differ in architecture, parameter count (100,676 against 178,068) and optimisation behaviour

as well as in normalisation, so a difference between them cannot be assigned to normalisation alone, and no mechanism is attributed.

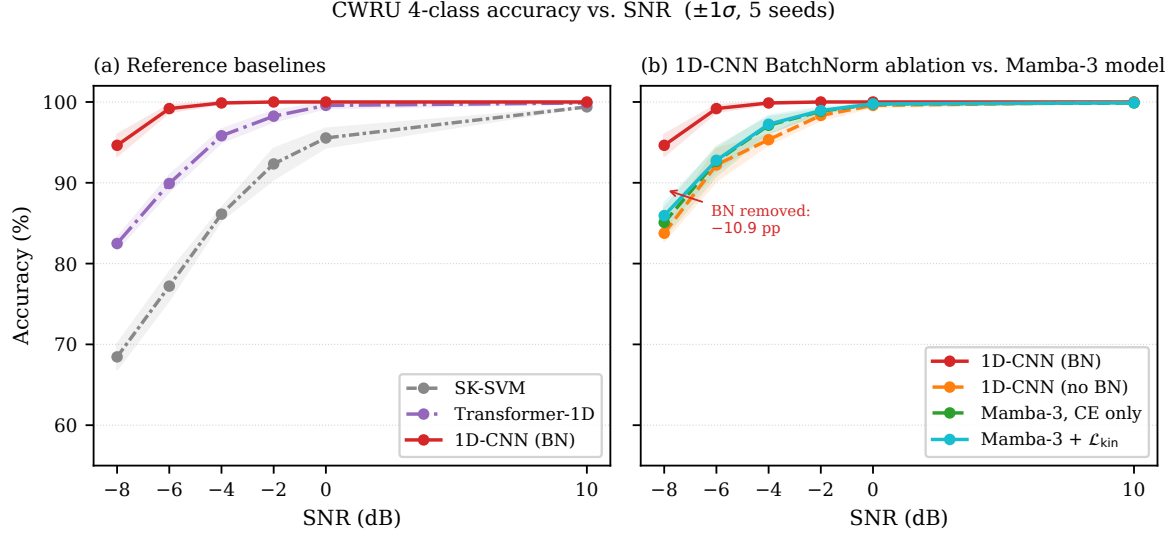


Figure A4. CWRU four-class DE accuracy against SNR (± 1 SD, 5 seeds). The rightmost point of every curve, plotted at the 10 dB tick, is the no-noise (clean) run. (a) Reference baselines: SK-SVM, Transformer-1D and 1D-CNN (BN). (b) Removing all eight BatchNorm layers lowers the 1D-CNN from 94.64% to 83.76% at -8 dB ($\Delta = -10.88$ pp); the Mamba-3 model, CE only (85.09%) and $+\mathcal{L}_{kin}$ (85.94%), overlaps the no-BN curve ($p = 0.19$). Replacing BatchNorm with `GroupNorm(1, d)` restores the within-condition accuracy (96.09% at -8 dB, table A12) but not the cross-condition result (table A13).

Within condition (CWRU). The 1D-CNN (BN) reaches $94.64 \pm 1.31\%$ at -8 dB, 8.7 pp above the Mamba-3 model with $\mathcal{L}_{kin}$ ($85.94 \pm 1.54\%$; table A7). To examine whether this difference depends on BatchNorm, the eight BatchNorm layers of the 1D-CNN were removed (no normalisation; convolution, GELU and max-pooling retained; table A10).

GroupNorm replacement. A criterion was fixed in advance in the experiment log: all eight `BatchNorm1d` layers of the 1D-CNN were replaced with `nn.GroupNorm(1, d)` (`baselines/cnn1d_ln.py`; architecture and hyperparameters otherwise unchanged). For each sample, `GroupNorm(1, d)` normalises jointly over all d channels and all time steps of a feature map and then applies a learned per-channel scale and shift; it uses no batch statistics. It is a layer-type normalisation but is not identical to a per-token LayerNorm, which would normalise over the channels separately at each time step. The experiment log and the file name call this variant “LN”; it is called `GroupNorm(1, d)` here. The cross-condition recovery

$$\text{recovery} = \frac{\text{CNN-GN}_{\text{cross}} - \text{CNN-BN}_{\text{cross}}}{\text{Mamba3}_{\text{cross}} - \text{CNN-BN}_{\text{cross}}}$$

Table A10. BatchNorm ablation: 1D-CNN with and without BatchNorm and the Mamba-3 model, CWRU four-class DE, leakage-free protocol (% , mean $\pm$ SD, 5 seeds). The no-BN 1D-CNN removes all eight `BatchNorm1d` layers; all other hyperparameters (learning rate 3×10^{-4} , 50 epochs, AdamW) are unchanged.

Learner	Clean	−4 dB	−6 dB	−8 dB
1D-CNN (with BN)	100.00 $\pm$ 0.00	99.88 $\pm$ 0.13	99.18 $\pm$ 0.44	94.64 $\pm$ 1.31
1D-CNN (no BN)	99.97 $\pm$ 0.07	95.33 $\pm$ 0.59	92.21 $\pm$ 1.99	83.76 $\pm$ 0.94 ^a
Mamba-3 model, CE only	99.88 $\pm$ 0.13	97.12 $\pm$ 0.99	92.67 $\pm$ 1.79	85.09 $\pm$ 1.76
Mamba-3 model, + $\mathcal{L}_{\text{kin}}$	99.91 $\pm$ 0.08	97.24 $\pm$ 1.07	92.79 $\pm$ 1.38	85.94 $\pm$ 1.54

^a Removing BatchNorm lowers 1D-CNN accuracy by 10.88 pp at −8 dB (two-sided Wilcoxon: 5/5 seeds concordant, $p = 0.063$, the $n = 5$ floor). Without BatchNorm the 1D-CNN (83.76%) is not significantly different from the Mamba-3 model (85.09%, $p = 0.19$). The no-BN variant was not re-tuned, so its accuracy may be a conservative lower bound. Adding BatchNorm to the Mamba-3 model lowered its −8 dB accuracy by 2.57 pp (table A11); the reason was not investigated.

Table A11. Effect of adding BatchNorm to the Mamba-3 model (after the convolutional embedding; +128 parameters, 0.07% of the total), CWRU DE, leakage-free protocol, mean $\pm$ SD over 5 seeds.

Variant	Clean	−4 dB	−8 dB
Mamba-3 model (no BN, original)	99.88 $\pm$ 0.13	97.12 $\pm$ 0.99	85.09 $\pm$ 1.76
Mamba-3 model + BN (ablation)	99.97 $\pm$ 0.07	95.33 $\pm$ 1.23	82.52 $\pm$ 1.05
Δ (BN added)	+0.09	−1.79	−2.57

Adding BN lowered accuracy at −4 and −8 dB in this configuration ($n = 5$); no explanation was tested.

was to be read as indicating a BatchNorm-specific failure if ≥ 0.5 , and as the normalisation swap not closing the difference in this setup if < 0.5 . Within condition (table A12), GroupNorm(1, d) matched BatchNorm closely across the SNR grid (largest difference at −8 dB: 96.09% with GroupNorm against 94.64% with BN, GroupNorm slightly higher). This addresses one possible confound of the no-normalisation ablation: the 10.88 pp drop could have reflected training without any normalisation rather than the loss of batch statistics. The GroupNorm result is consistent with the within-condition difference depending on having a normalisation layer rather than on batch statistics specifically, for this CNN on this dataset.

Table A12. 1D-CNN with GroupNorm(1, d) (GN) and with BatchNorm (BN), CWRU within-condition DE accuracy, leakage-free protocol (% , mean $\pm$ SD, 5 seeds).

Learner	Clean	0 dB	−2 dB	−4 dB	−6 dB	−8 dB
1D-CNN (BN)	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.88 $\pm$ 0.13	99.18 $\pm$ 0.44	94.64 $\pm$ 1.31
1D-CNN (GN)	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	99.64 $\pm$ 0.25	99.76 $\pm$ 0.14	98.82 $\pm$ 0.54	96.09 $\pm$ 0.62

Across conditions (XJTU-SY condition 2→3, H), the GroupNorm(1, d) 1D-CNN reached $42.72 \pm 0.97\%$ macro-F1 ($n = 5$, leakage-free), against 40.23% for BN and 63.09% for the Mamba-3 model ($n = 8$, supplementary table S14): a recovery of $(42.72 - 40.23)/(63.09 - 40.23) = 0.109$, below the 0.5 criterion. Replacing the normalisation layer did not close the cross-condition difference in this setup. The ratio mixes an $n = 5$ GroupNorm(1, d) arm with $n = 8$ BatchNorm and Mamba-3 arms. For comparison, adding V to the 1D-CNN (supplementary table S14) raised its macro-F1 from 40.23% to 45.50%, a ratio of $(45.50 - 40.23)/(63.09 - 40.23) = 0.231$. Neither change closes much of the difference (10.9% by normalisation replacement, 23.1% by the added channel); both 1D-CNN variants stay near the constant-predictor floor, so the ratios are small differences near that floor and no conclusion is drawn from their ordering.

Across conditions. Table A13 compares the Mamba-3 model and the 1D-CNN on XJTU-SY condition 2→3 with H input (+7% shaft-speed shift from 2250 to 2400 rpm).

Table A13. XJTU-SY condition 2→3, H input, first software environment, leakage-free protocol (fixed final epoch): macro-F1 (%) and per-class recall. The 1D-CNN (with BN) uses the same pipeline as the Mamba-3 model; both use weighted random sampling (OR:IR = 7.4:1). 1D-CNN and Mamba-3 CE-only rows: $n = 8$ seeds (supplementary section S6.2); Mamba-3 + $\mathcal{L}_{\text{kin}}$ row: $n = 5$ (not re-run).

Learner	macro-F1 (%)	OR recall (%)	IR recall (%)	Δ macro-F1
1D-CNN (BN)	40.23 ± 0.16	100.0	0.2 ± 0.1	—
Mamba-3 model, CE only	63.09 ± 8.42	100.0	27.1 ± 11.6	+22.9 pp
Mamba-3 model, + $\mathcal{L}_{\text{kin}}$	61.89 ± 8.41	100.0	25.4 ± 12.2	+21.7 pp

The 1D-CNN is essentially a constant-OR predictor (IR recall 0.2%, 8/8 seeds) despite weighted random sampling; the GroupNorm(1, d) variant behaves similarly (macro-F1 42.72%), so the failure is not specific to BatchNorm. Why it fails was not tested; the +7% speed shift changes fault-frequency positions and amplitude envelopes, and these effects were not separated. The Mamba-3 model also has OR recall 100% but retains partial IR recall (27.1% on average; per-seed range 9–44% CE only, 13–45% with $\mathcal{L}_{\text{kin}}$), so its predictions are not constant; the difference from the 1D-CNN lies in the IR recall. CE only and + $\mathcal{L}_{\text{kin}}$ are on par (63.09 against 61.89 macro-F1), so the difference from the 1D-CNN is not attributable to the regulariser. 1D-CNN (BN) against Mamba-3 CE only: two-sided Wilcoxon $p = 0.0078$ (the $n = 8$ floor), 8/8 seeds concordant. The + $\mathcal{L}_{\text{kin}}$ Δ is a mean difference against the $n = 5$ subset of the 1D-CNN baseline, not a paired $n = 8$ test. All of this is one condition pair.

A2.5. Summary of the learner comparisons

- On CWRU under AWGN (DE and DE+FE), the BatchNorm 1D-CNN and the Mamba-2 model scored higher than the Mamba-3 model at low SNR (tables A7, A16 and A9); without BatchNorm the 1D-CNN was not significantly different from the

Mamba-3 model (83.76% against 85.09%, $p = 0.19$, $n = 5$), and GroupNorm(1, d) restored its accuracy.

- On Paderborn cross-condition, no difference between the Mamba-3 and Mamba-2 models was detected (table A8).
- On XJTU-SY condition 2→3 the BatchNorm 1D-CNN was a near-constant OR predictor (IR recall 0.2%), the Mamba-3 model retained 63.1% macro-F1 with H input ($n = 8$, $p = 0.0078$ against the 1D-CNN), and the GroupNorm(1, d) 1D-CNN stayed near the same floor (table A13).

These are observations from three datasets with learners that differ in architecture and size. No account of them in terms of optimisation behaviour, source-domain statistics or inductive bias was tested, no within-condition versus cross-condition selection rule is claimed, and no learner is claimed to be generally more accurate. The sensor-configuration findings of the main text are stated per learner and do not depend on these comparisons.

A3. Forward-pass latency microbenchmark

A forward-only latency benchmark, without training, compared the Mamba-3 model (single-input single-output, SISO), the Mamba-3 model in MIMO mode (rank 4, forward only), the Mamba-2 model, the 1D-CNN and a Transformer-1D on a single 2048-sample window, one sensor channel, default hyperparameters ($d_{\text{model}} = 64$, 4 layers). Each model was warmed up for 10 iterations and then timed over 50 forward passes (`torch.cuda.synchronize()` before and after each call) on one GPU (RTX A5000 laptop, Ampere sm.86). This is a single measured configuration. The window length is fixed and only the batch size varies, so the benchmark does not test how latency scales with sequence length and cannot verify linear-time complexity; the linear cost in sequence length of the state-space backbones is a property stated by the backbone authors, not a result measured here. The MIMO row is forward-only, because backward passes for MIMO rank ≥ 2 could not be run on this GPU (supplementary section S7.3).

Table A14. Forward-pass latency (ms, mean $\pm$ SD over 50 timed calls), window length 2048, one sensor channel.

Model	Parameters	batch size 1 (ms)	batch size 64 (ms)
1D-CNN	100,676	0.98 ± 0.12	2.81 ± 0.15
Transformer-1D	200,900	1.53 ± 0.06	68.34 ± 2.34
Mamba-2 model	173,980	4.06 ± 0.20	13.18 ± 0.08
Mamba-3 model (SISO)	178,068	5.43 ± 1.30	17.69 ± 1.07
Mamba-3 model (MIMO) ^a	386,964	4.45 ± 0.85	27.31 ± 1.41

^a MIMO rank $r = 4$; forward only (not trainable on this GPU; supplementary section S7.3).

From batch size 1 to 64 the Transformer-1D latency increased by about $45\times$, and that of the state-space models and the 1D-CNN by $3\text{--}6\times$. This reflects how each model uses the GPU at a fixed window length and does not by itself show a dependence on sequence length. The Mamba-3 model (SISO) was slower than the Mamba-2 model at both batch sizes (5.43 against 4.06 ms at batch size 1; 17.69 against 13.18 ms at batch size 64). The MIMO rank-4 variant has more than twice the parameters and was about $1.5\times$ slower than SISO at batch size 64. Script: `experiments/step5_throughput_bench.py`; raw results: `results/step5_throughput_bench_20260721/results.json` (repository paths).

A4. Exploratory kinematic frequency-alignment regulariser (not a contribution)

Status. This regulariser is reported for completeness. Under the leakage-free protocol it gave no consistent gain in accuracy or inter-seed variance on CWRU, Paderborn or XJTU-SY, and an earlier claim that it improved out-of-distribution accuracy was withdrawn (the model without it reaches 63.09% single-sensor cross-condition macro-F1, with it 61.89%; table A13). Nothing in the main manuscript depends on it. The text below describes the method and reports that the learned state frequencies align with fault harmonics; that alignment is a property of the state space and is not evidence of better diagnosis.

A4.1. Bearing fault characteristic frequencies

For a bearing with n_b rolling elements, ball diameter d , pitch diameter D and contact angle α , the four primary fault characteristic frequencies are [3, 4]

$$f_{\text{BPFO}} = \frac{n_b}{2} f_r \left(1 - \frac{d}{D} \cos \alpha \right), \quad (\text{A1})$$

$$f_{\text{BPFI}} = \frac{n_b}{2} f_r \left(1 + \frac{d}{D} \cos \alpha \right), \quad (\text{A2})$$

$$f_{\text{BSF}} = \frac{D}{2d} f_r \left[1 - \left(\frac{d}{D} \cos \alpha \right)^2 \right], \quad (\text{A3})$$

$$f_{\text{FTF}} = \frac{f_r}{2} \left(1 - \frac{d}{D} \cos \alpha \right), \quad (\text{A4})$$

where $f_r = \Omega/60$ is the shaft rotation frequency (Hz) and Ω the shaft speed (rpm). f_{BPFO} (ball pass frequency, outer race) is excited by outer-race defects, f_{BPFI} by inner-race defects, f_{BSF} (ball spin frequency) by rolling-element defects and f_{FTF} (fundamental train frequency) by cage defects. Each frequency and its harmonics produce periodic impulse trains in the measured vibration, which are the classical diagnostic signatures [5, 2]. Table A15 lists the bearing parameters and the resulting frequencies; the same table is used by the coherence analysis (supplementary section S1.2).

Table A15. Bearing parameters and fault characteristic frequencies at nominal shaft speed. All contact angles $\alpha = 0^\circ$.

Dataset	Bearing	n_b	d (mm)	D (mm)	Ω (rpm)	$f_{\text{BPFO}} / f_{\text{BPFI}}$ (Hz)
CWRU DE	SKF 6205	9	7.94	39.04	1797	107.36 / 162.19
CWRU FE	SKF 6203	8	6.75	28.50	1797	91.44 ^a / 148.16 ^a
Paderborn	SKF 6203	8	6.75	29.05	1500	76.77 / 123.23
XJTU-SY	LDK UER204	8	7.92	34.55	2400	123.32 / 196.68

^a The CWRU FE bearing (SKF 6203) has a different pitch diameter (28.50 mm) from the Paderborn bearing of the same part number (29.05 mm); the two are computed separately.

A4.2. Per-sample frequency targets

All characteristic frequencies scale with f_r , so a change of operating speed shifts the whole fault spectrum, and a fixed dataset-level target set would be misaligned by an amount proportional to the speed deviation [6, 7]. The target set $\mathcal{F}_i$ is therefore computed per sample from the sample’s shaft speed Ω_i , which is available as metadata in all three datasets (CWRU nominal 1797 rpm; Paderborn 900 or 1500 rpm; XJTU-SY 2100, 2250 or 2400 rpm per condition). The speed metadata are nominal and constant within a condition; within-window speed traces are not available and speed fluctuation inside a window is not modelled. In these experiments the speed is a per-condition constant (three values for XJTU-SY, two for Paderborn, one for CWRU), so continuously varying speed was not tested. Spectral kurtosis and envelope demodulation [2] are the classical signal-processing route to fault-frequency content; the regulariser was designed to pull the internal oscillatory states of the Mamba-3 model toward the analytically known fault harmonics during training. It was not compared with envelope analysis.

A4.3. Definition of the regulariser

Instantaneous state frequency. The official Mamba-3 kernel [8] computes an angle activation $\theta_{h,k}(t) \in \mathbb{R}$ for each head h and rotary dimension k at every time step t . Following the phase-accumulation formula of the kernel source (`ops/triton/mamba3/angle_dt.py`, lines 94–104), the instantaneous frequency assigned to state (h, k) at step t is

$$f_{h,k}(t) = \tanh(\theta_{h,k}(t)) \cdot \Delta t_h(t) \cdot \frac{f_s^{\text{eff}}}{2} \quad [\text{Hz}], \quad (\text{A5})$$

where $\Delta t_h(t) > 0$ is the learned time step of head h and $f_s^{\text{eff}} = f_s/s_c$ the effective sampling rate after the convolutional stride. Because $|\tanh(\theta)| < 1$, $|f_{h,k}(t)|$ is bounded by $\Delta t_h(t) f_s^{\text{eff}}/2$. This equals the Nyquist limit $f_s^{\text{eff}}/2$ only when $\Delta t_h(t) \leq 1$; the constraint $\Delta t_h(t) > 0$ alone does not imply it, and $\Delta t_h(t) \leq 1$ is not enforced. Where $\Delta t_h(t) > 1$, the quantity in equation (A5) is a per-step phase increment expressed in

Hz and is defined only modulo f_s^{eff} (it can alias). $f_{h,k}(t)$ is therefore read as an internal rotation rate, not as a frequency guaranteed to lie in the observable band; how often $\Delta t_h(t)$ exceeds one was not measured. The time-averaged frequency of each state is $\bar{f}_{h,k} = \frac{1}{L} \sum_t f_{h,k}(t)$; these values are stored as the state-frequency snapshot $\bar{\mathbf{f}} \in \mathbb{R}^{B \times S}$, with $S = N_\ell \times S_{\text{rope}} = 4 \times 64 = 256$ rotary states over all layers (32 rotary pairs per head, 2 heads per layer).

Fault frequency targets. For each sample i with shaft speed Ω_i ,

$$\mathcal{F}_i = \{f_r^i, f_{\text{FTF}}^i, f_{\text{BPFO}}^i, f_{\text{BSF}}^i, f_{\text{BPFI}}^i\} \cup \text{2nd and 3rd harmonics}, \quad (\text{A6})$$

which gives 15 target frequencies. Ω_i is the nominal speed of the sample’s operating condition, so the target set is constant within a condition.

Cover variant. The cover variant requires that every target $f_c \in \mathcal{F}_i$ be matched by at least one state. With the nearest-state distance $\delta_{c,i} = \min_{(h,k)} |f_{h,k}(t) - f_c|$, aggregated over time,

$$\mathcal{L}_{\text{kin}} = \frac{1}{B} \sum_{i=1}^B \frac{1}{|\mathcal{F}_i|} \sum_{f_c \in \mathcal{F}_i} \frac{1}{L} \sum_{t=1}^L \min_{(h,k)} |f_{h,k}(t) - f_c|, \quad (\text{A7})$$

where B is the batch size. The total loss is $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{kin}}$ with $\lambda = 0.01$ (table A17). $\mathcal{L}_{\text{kin}}$ is computed in float32 even when training in bf16, to avoid gradient underflow in the inner minimum.

Applicability. Equation (A5) uses the angle activations of Mamba-3’s complex-state rotation. Real-valued state-space models such as Mamba-2 have no instantaneous rotation frequencies, so $f_{h,k}(t)$ is undefined for them, and among the learners used here only the Mamba-3 model can carry $\mathcal{L}_{\text{kin}}$. This is a structural property, not a result.

Relation to normalisation. $\mathcal{L}_{\text{kin}}$ is an auxiliary objective whose targets are derived analytically from shaft speed. BatchNorm normalises activations with source-domain batch statistics, which has been reported to be sensitive to domain shift [9, 10]. In one CWRU experiment, the Mamba-3 model with BatchNorm after the convolutional embedding scored 2.6 pp lower at -8 dB than without it (table A11, $n = 5$); this is a single-configuration observation, and an explanation in terms of an interaction between batch normalisation and complex-state parameters was not tested. Whether $\mathcal{L}_{\text{kin}}$ is more robust to domain shift than normalisation was not tested either; the cross-condition differences between the Mamba-3 model and the 1D-CNN (table A13) are CE-only results and are not attributable to the regulariser.

A4.4. Results under the leakage-free protocol

Test direction. The comparisons of $+\mathcal{L}_{\text{kin}}$ with CE only on CWRU (four-class SNR grid, table A16; ten-class, table A18) were tested one-sided (exact Wilcoxon signed-

rank test, alternative: $+\mathcal{L}_{\text{kin}}$ more accurate than CE only, $n = 5$), as originally planned. The main text declares only one directional hypothesis, which concerns the sensors (section 3.4); these one-sided regulariser tests are reported here as they were run. Because the exact Wilcoxon statistic is symmetric under the null hypothesis, the two-sided p equals $\min(1, 2p)$ when the observed statistic lies in the direction of the one-sided alternative (one-sided $p \leq 0.5$); it is given in square brackets next to each such value. Values marked $\dagger$ lie in the opposite direction, where $\min(1, 2p)$ does not give the two-sided value, and no two-sided value is given. The Paderborn comparisons (table A19) were two-sided. No correction for multiple comparisons was applied.

CWRU, accuracy and inter-seed SD. Table A16 and figure A5 give the accuracy of the Mamba-3 model with and without $\mathcal{L}_{\text{kin}}$, and of the Mamba-2 model, over the SNR grid (DE). $\mathcal{L}_{\text{kin}}$ changed the mean accuracy by at most 0.85 pp (at -8 dB) and changed the inter-seed SD in both directions across the grid. An earlier version of this study, using window-level random splits, had reported a 14–62% reduction of the SD; that effect did not survive the split correction and is withdrawn.

Table A16. CWRU four-class DE test accuracy (%) against SNR, leakage-free protocol (mean $\pm$ SD, 5 seeds, first software environment).

SNR (dB)	Mamba-2, CE only	Mamba-3, CE only	Mamba-3, $+\mathcal{L}_{\text{kin}}$
+10	99.97 $\pm$ 0.07	99.94 $\pm$ 0.14	99.91 $\pm$ 0.08
0	99.67 $\pm$ 0.59	99.76 $\pm$ 0.08	99.76 $\pm$ 0.08
-2	99.55 $\pm$ 0.41	98.85 $\pm$ 0.23	98.94 $\pm$ 0.40
-4	98.21 $\pm$ 1.13	97.12 $\pm$ 0.99	97.24 $\pm$ 1.07
-6	95.88 $\pm$ 0.97	92.67 $\pm$ 1.79	92.79 $\pm$ 1.38
-8	92.03 $\pm$ 1.10	85.09 $\pm$ 1.76	85.94 $\pm$ 1.54

$+\mathcal{L}_{\text{kin}}$ minus CE only, mean difference $-0.03/0.00/+0.09/+0.12/+0.12/+0.85$ pp, and SD ratio ($+\mathcal{L}_{\text{kin}}$ /CE only) $0.61/1.00/1.74/1.08/0.77/0.88$ (+10 to -8 dB): neither quantity moves in a consistent direction. One-sided exact Wilcoxon p ($+\mathcal{L}_{\text{kin}} >$ CE only, $n = 5$, as originally planned), +10 to -8 dB: $0.6250^\dagger$, $0.6875^\dagger$, 0.3125 [0.6250], 0.4375 [0.8750], $0.5938^\dagger$, 0.0625 [0.1250]; square brackets: two-sided $\min(1, 2p)$; $\dagger$: observed direction opposite to the alternative (per-cell values from the repository record `experiments/exp_bm3_cwru_leakfree/EVIDENCE.REPORT_bm3_fix_addendum2.md`; earlier versions reported only the bound $p \geq 0.0625$). No improvement is claimed. Mamba-2 against Mamba-3 CE only (two-sided): $p = 0.1875/0.0625/0.0625$ at $-4/-6/-8$ dB (section A2.3).

Weight λ and ten-class setting. Table A17 reports accuracy at SNR = 0 dB for $\lambda \in \{10^{-3}, 10^{-2}, 10^{-1}, 1\}$. All four values give comparable accuracy (range 0.06 pp), so accuracy is insensitive to λ in this range; $\lambda = 0.01$ is used throughout. In a ten-class CWRU setting (normal plus IR, ball and OR at three fault diameters, SNR = 0 dB; table A18), CE only and $+\mathcal{L}_{\text{kin}}$ reached $97.88 \pm 0.50\%$ and $97.85 \pm 0.39\%$ (one-sided $p = 0.6875^\dagger$): no effect on the mean, and a SD difference within seed noise.

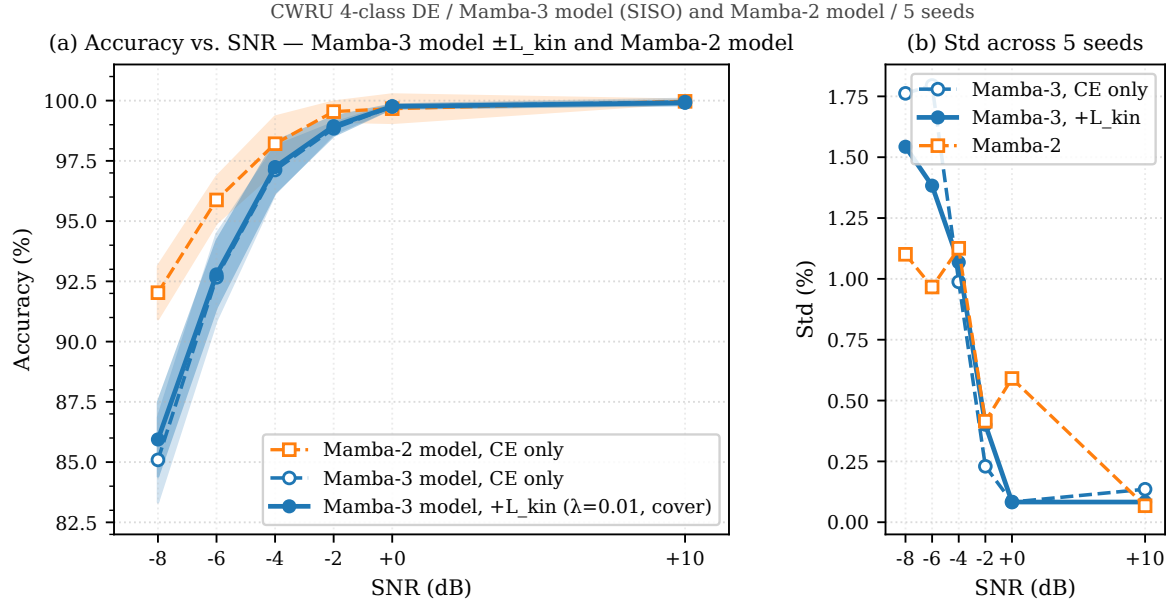


Figure A5. CWRU four-class DE test accuracy (a) and inter-seed SD (b) against SNR, leakage-free protocol, for the Mamba-2 model (CE only) and the Mamba-3 model, CE only and $+\mathcal{L}_{\text{kin}}$ ($\lambda = 0.01$, 5 seeds). (a) The Mamba-2 model is higher at -6 and -8 dB ($p = 0.063$, two-sided) and marginally at -4 dB ($p = 0.19$); $\mathcal{L}_{\text{kin}}$ neither shifts the mean curve nor consistently narrows the SD band relative to CE only. (b) The inter-seed SD with $\mathcal{L}_{\text{kin}}$ changes in both directions across the SNR grid (table A16).

Table A17. CWRU four-class DE test accuracy (%) against λ at SNR = 0 dB, Mamba-3 model, leakage-free protocol, mean $\pm$ SD (5 seeds).

λ	Accuracy (%)	Δ vs CE only (pp)
CE only ($\lambda = 0$)	99.76 ± 0.08	—
10^{-3}	99.76 ± 0.14	0.00
10^{-2} (default)	99.76 ± 0.14	0.00
10^{-1}	99.70 ± 0.11	−0.06
1.0	99.70 ± 0.15	−0.06

Accuracy range across $\lambda \in [10^{-3}, 1]$: 0.06 pp; accuracy is insensitive to λ in this range. $\lambda = 0.01$ is used in all other experiments.

Paderborn. Paderborn switches between two discrete operating speeds (900 and 1500 rpm), so the per-sample speed target of $\mathcal{L}_{\text{kin}}$ reduces to a two-point set. Table A19 gives the accuracy of the Mamba-3 model with and without $\mathcal{L}_{\text{kin}}$ under AWGN. No mean difference is significant at any noise level ($p \geq 0.13$), and the sign of the difference is not stable across noise levels. The discrete speed regime is one possible reason, which was not tested. Selecting on the last epoch instead of the best validation epoch changes no test accuracy by more than 0.31 pp. The original and leakage-free values are compared in supplementary table S13.

Table A18. CWRU ten-class DE test accuracy (%) at SNR = 0 dB, Mamba-3 model, leakage-free protocol, mean $\pm$ SD (5 seeds). The ten-class setting includes normal plus IR, ball and OR at 0.007, 0.014 and 0.021 inch (one class per fault type and diameter; label map LABEL10 in `bearmamba3/data_cwru.py`).

Training	Accuracy (%)	SD (%)
CE only	97.88 $\pm$ 0.50	0.50
+ $\mathcal{L}_{\text{kin}}$ ($\lambda = 0.01$)	97.85 $\pm$ 0.39	0.39
Δ mean (pp)	−0.03 (one-sided $p = 0.6875^\dagger$)	
SD ratio (+ $\mathcal{L}_{\text{kin}}$ /CE only)	0.78	

No effect on the mean; SD difference within seed noise. One-sided exact Wilcoxon (+ $\mathcal{L}_{\text{kin}}$ > CE only), as originally planned; $\dagger$: observed direction opposite to the alternative, so no two-sided value is derived.

Table A19. Paderborn cross-condition accuracy (%) under AWGN, Mamba-3 model, leakage-free protocol (source-domain validation split; test condition evaluated once), mean $\pm$ SD (5 seeds). p : two-sided exact Wilcoxon, + $\mathcal{L}_{\text{kin}}$ against CE only.

SNR	CE only	+ $\mathcal{L}_{\text{kin}}$	Δ (pp)	p
Clean	95.19 $\pm$ 0.83	95.49 $\pm$ 0.76	+0.30	0.125
0 dB	88.31 $\pm$ 0.71	88.57 $\pm$ 0.57	+0.26	1.000
−4 dB	83.35 $\pm$ 0.90	83.09 $\pm$ 0.41	−0.26	0.4375

XJTU-SY. With H input on condition 2→3, CE only reached 63.09% and + $\mathcal{L}_{\text{kin}}$ 61.89% macro-F1 (table A13); with H+V both reached 89.72%, and under LOBO the two training arms differ by 0.05 pp (supplementary tables S18 and S19). The accuracy-side outcome of $\mathcal{L}_{\text{kin}}$ is therefore negative on all three datasets. The continuous-speed regime, for which the per-sample targets were designed, was not tested.

A4.5. Alignment of state frequencies with fault harmonics

This subsection reports only that, with $\mathcal{L}_{\text{kin}}$, the learned state frequencies moved toward the fault harmonics during training. This describes the state space; it is not evidence of better diagnosis, and it did not translate into accuracy on any dataset (section A4.4).

State-frequency snapshot. During training, the time-averaged state frequencies $\bar{\mathbf{f}} \in \mathbb{R}^{B \times 256}$ were recorded and compared with the per-sample targets $\mathcal{F}_i$. On CWRU at SNR = 0 dB (figure A6; 50 epochs, 5 seeds), the density of state frequencies concentrated around the fault harmonics and the mean nearest-state distance fell by $5.6\times$ (BPFO) and $2.5\times$ (BPFI). On XJTU-SY condition 2→3 (figure A7) the distance fell faster (BPFI: $11.3\times$ by epoch 10). A possible reason, not tested, is that the large-amplitude run-to-failure impulses of XJTU-SY already populate the fault band from the first epoch.

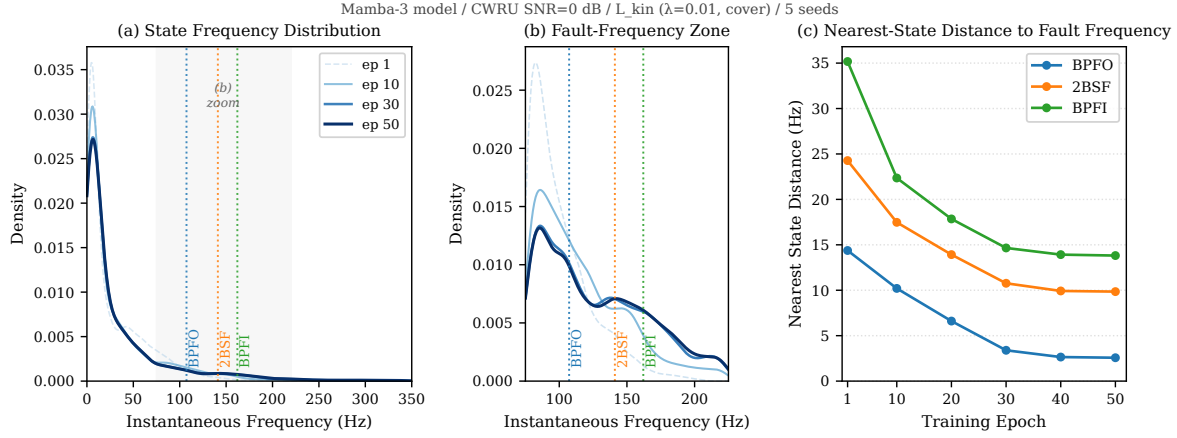


Figure A6. State-frequency snapshot, Mamba-3 model, CWRU DE, SNR = 0 dB, $+\mathcal{L}_{\text{kin}}$ ($\lambda = 0.01$, cover), 5 seeds (640 samples per epoch). (a) Kernel density of all time-averaged state frequencies $\bar{f}_{h,k}$ in 0–300 Hz at epochs 1 (lightest blue) to 50 (darkest blue); the yellow band marks the 60–230 Hz fault-frequency zone. (b) Zoom on the fault-frequency zone; dashed lines mark BPFO (107.4 Hz), 2BSF (141.2 Hz) and BPFI (162.2 Hz). (c) Mean nearest-state distance against epoch: BPFO 14.4 $\rightarrow$ 2.6 Hz (5.6 $\times$); BPFI 35.2 $\rightarrow$ 13.8 Hz (2.5 $\times$).

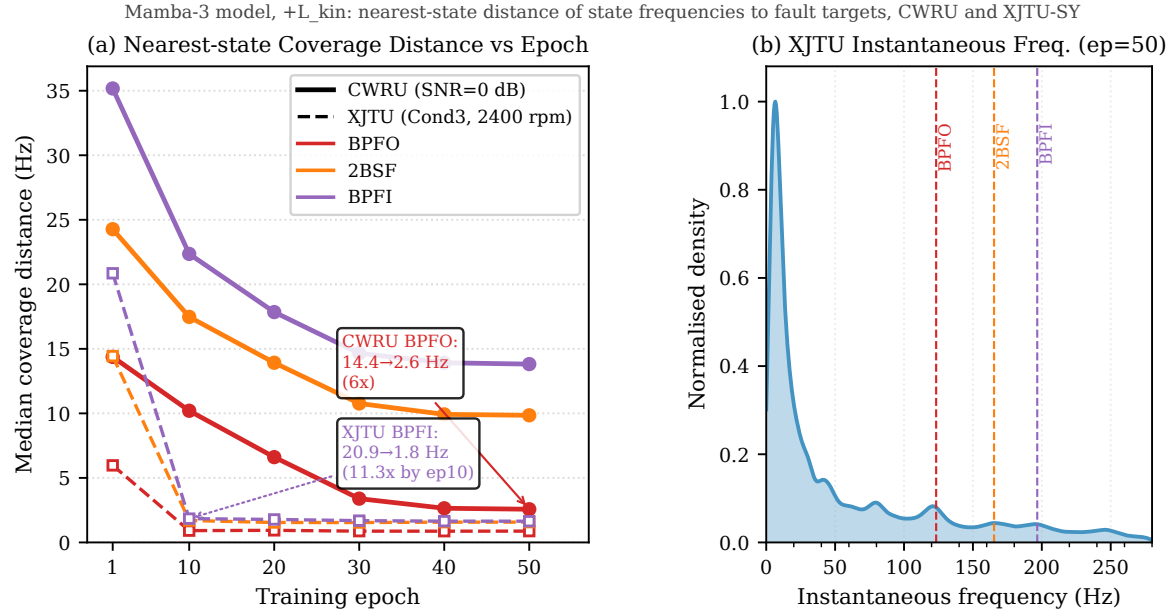


Figure A7. Nearest-state distance on CWRU DE (SNR = 0 dB, solid lines, circles) and XJTU-SY condition 2 $\rightarrow$ 3 (dashed lines, squares), both $+\mathcal{L}_{\text{kin}}$, 5 seeds, leakage-free training runs, per-sample targets. (a) Mean nearest-state distance (Hz) to BPFO (red), 2BSF (orange) and BPFI (blue) over 50 epochs. On XJTU-SY the BPFI distance drops from 20.9 to 1.8 Hz (11.3 $\times$) by epoch 10, against 35.2 $\rightarrow$ 13.8 Hz (2.5 $\times$) for CWRU over 50 epochs. (b) State-frequency density for XJTU-SY at epoch 50 in the 80–280 Hz zone; dashed lines mark the 2400 rpm fault harmonics (BPFO = 123.3 Hz, 2BSF = 165.3 Hz, BPFI = 196.7 Hz). The state frequencies align with fault harmonics on both datasets.

Clustering of state frequencies and of pre-classifier features. In the 256-dimensional state-frequency space (figure A8) the Davies–Bouldin index of the fault classes decreased from 15.08 at epoch 1 to 10.90 at epoch 50 (-27.8%). In the pre-classifier feature space (figure A9) the classes are already well separated without $\mathcal{L}_{\text{kin}}$ at SNR = 0 dB (DB 0.396 against 0.395 with $\mathcal{L}_{\text{kin}}$, $\Delta < 1\%$, sign not stable across seeds), and at -6 dB they overlap heavily with and without $\mathcal{L}_{\text{kin}}$ (DB 0.864 against 0.868, within the inter-seed range; section A5). The regulariser acts on the state-frequency activations, not on the representation passed to the classifier, and it changed neither mean accuracy nor inter-seed variance under the leakage-free protocol. Why the alignment did not translate into accuracy is not established here. Possible reasons, none of them tested, include the near-ceiling CWRU accuracy at 0 dB, the two-point speed set of Paderborn, and the large-amplitude XJTU-SY impulses that reduce the coverage distance early in training; Jiang *et al* [11] discuss more generally that the effect of physical priors in diagnostic models depends on signal characteristics.

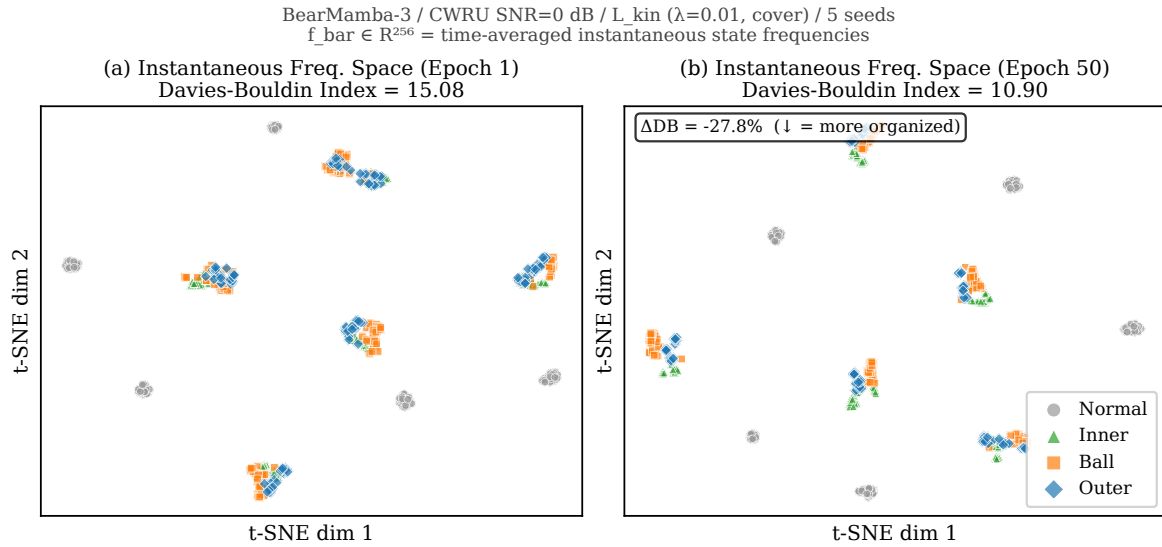


Figure A8. t-SNE of the time-averaged state frequencies $\bar{\mathbf{f}} \in \mathbb{R}^{256}$ at epoch 1 (a) and epoch 50 (b), Mamba-3 model, CWRU DE, SNR = 0 dB, $+\mathcal{L}_{\text{kin}}$ ($\lambda = 0.01$, cover), 5 seeds (320 samples plotted). Each point is a vibration window; colours denote fault classes. The Davies–Bouldin index decreases from 15.08 to 10.90 ($\Delta = -27.8\%$; lower is more compact). This describes the state-frequency space only; compare figure A9, which shows that the pre-classifier feature space is already well separated without $\mathcal{L}_{\text{kin}}$ at this SNR. The figure title uses the earlier name “BearMamba-3” for the Mamba-3 model, and the panel annotation’s “more organized” means a lower DB index (more compact clusters).

A4.6. Summary

Under the leakage-free protocol the kinematic regulariser gave no consistent change in accuracy or inter-seed variance on CWRU (four- and ten-class), Paderborn or XJTU-

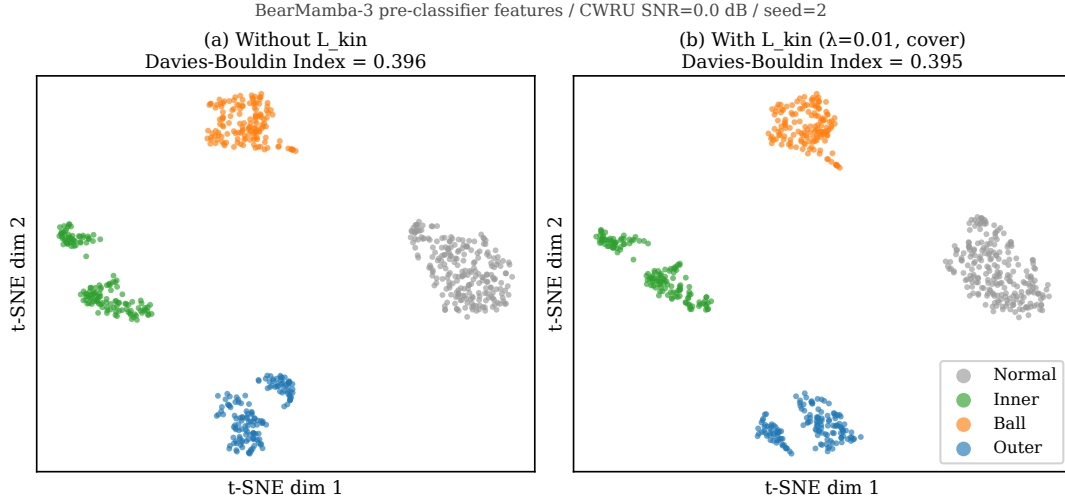


Figure A9. t-SNE of pre-classifier features, Mamba-3 model, CWRU DE, SNR = 0 dB, seed 2. (a) CE only: DB = 0.396, four well-separated clusters. (b) $+\mathcal{L}_{\text{kin}}$ ($\lambda = 0.01$): DB = 0.395, nearly identical cluster geometry. The pre-classifier feature space is near ceiling at this SNR, which is consistent with the non-significant difference in mean accuracy (one-sided $p = 0.6875^\dagger$ at 0 dB, table A16). The figure title uses the earlier name “BearMamba-3” for the Mamba-3 model.

SY, and the earlier claims of an out-of-distribution gain and of a variance reduction are withdrawn. Its only observed effect is that the learned state frequencies align with the fault harmonics, which is not evidence of better diagnosis. It is not a contribution of this study, and no result of the main text depends on it.

A5. Kinematic regulariser: pre-classifier feature-space figures

This section belongs to the exploratory kinematic regulariser of section A4, which gave no consistent gain in accuracy or inter-seed variance under the leakage-free protocol and is not a contribution of this study. Figures A10 and A11 show pre-classifier features of the CWRU DE Mamba-3 model with and without $\mathcal{L}_{\text{kin}}$, with the Davies–Bouldin (DB) index (lower values indicate more compact clusters).

The multi-seed check (figure A11) shows that seed 2, used for the single-seed figure, lies within the five-seed range at -6 dB rather than at an extreme. Neither figure is evidence that $\mathcal{L}_{\text{kin}}$ improves diagnosis.

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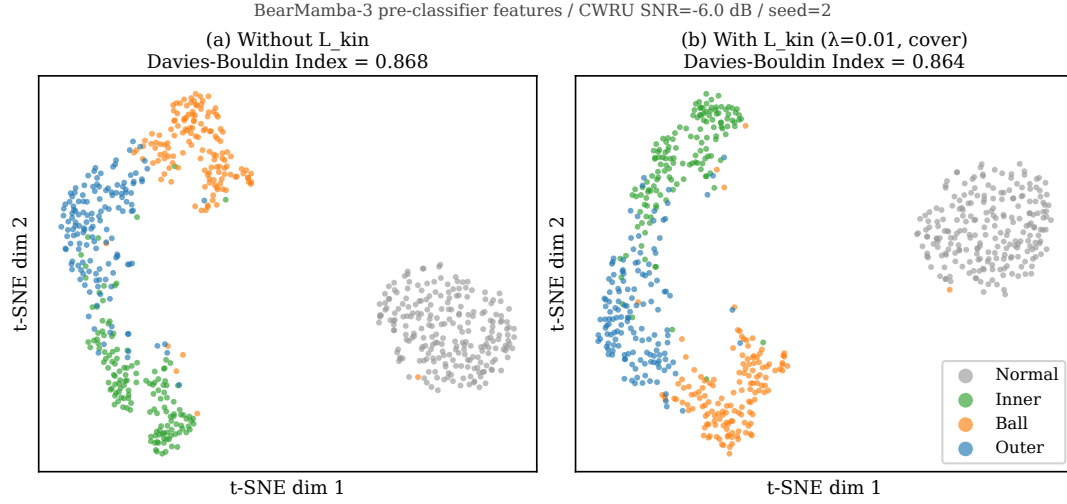


Figure A10. t-SNE of pre-classifier features, CWRU DE, Mamba-3 model, SNR = -6 dB, seed 2, leakage-free protocol. (a) CE only: DB = 0.868; (b) $+\mathcal{L}_{kin}$: DB = 0.864. At this noise level the fault-class clusters overlap substantially in both cases; $\mathcal{L}_{kin}$ does not recover separability. The two DB values differ by less than the inter-seed range and do not indicate a stable direction (figure A11). The figure title uses the earlier name “BearMamba-3” for the Mamba-3 model.

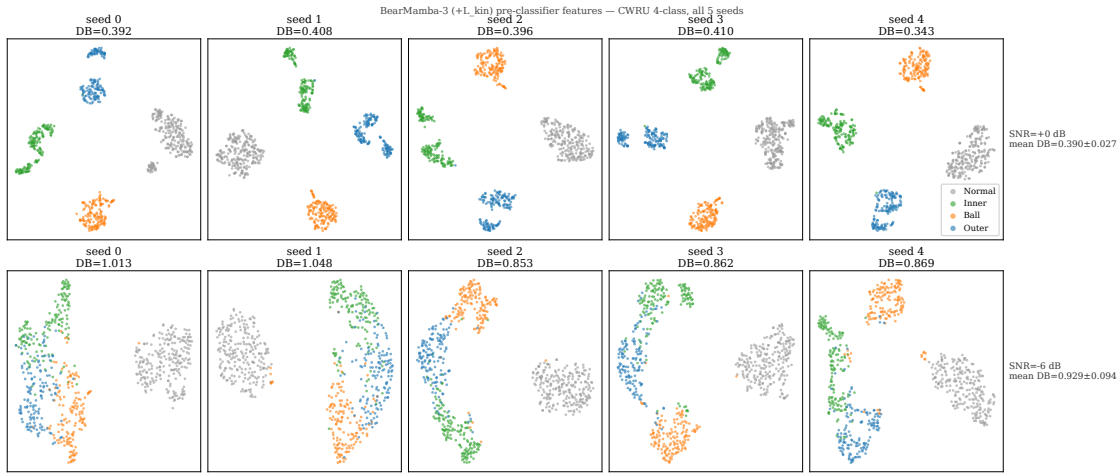


Figure A11. Cross-seed t-SNE check, leakage-free protocol: pre-classifier feature projections for all five training seeds at SNR = 0 dB (top row) and -6 dB (bottom row), DB index per seed. At 0 dB, DB = 0.390 ± 0.027 (5 seeds); at -6 dB, DB = 0.929 ± 0.094 , with the largest value for seed 1 (DB = 1.048). Excluding seed 1 leaves an SD of 0.076, still above the 0 dB value, so the larger dispersion at -6 dB is not due to one seed alone. Seed 2 (figure A10) has DB = 0.853 in this multi-seed batch, within the five-seed range; the standalone single-seed run of figure A10 gives 0.864 under an otherwise identical configuration, a small run-to-run difference that we report as observed. All seeds use the same t-SNE settings (perplexity = 30, random_state = 42). The figure title uses the earlier name “BearMamba-3” for the Mamba-3 model.

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